

34 The molecular formula of compound Z is C_4H_{10} .

Which row identifies the substance that reacts with Z and describes the type of reaction that occurs?

	substance	type of reaction
A	chlorine	addition
B	chlorine	substitution
C	steam	addition
D	steam	substitution

35 Which list shows the fractions obtained from the fractional distillation of petroleum, in order of increasing boiling point?

- A** bitumen → diesel oil → fuel oil → lubricating oil
B diesel oil → gasoline → naphtha → kerosene
C gasoline → naphtha → kerosene → diesel oil
D kerosene → lubricating oil → naphtha → refinery gas

36 Three statements about cracking of larger alkane molecules are listed.

- 1 Cracking produces petrol for cars.
- 2 Cracking only produces short-chain alkenes.
- 3 Cracking produces alkenes used to make polymers.

Which statements are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

37 Some words used to describe organic compounds are listed.

- 1 hydrocarbon
- 2 monomer
- 3 saturated
- 4 unreactive

Which words describe ethene?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4